

# PSY 39200CPAI Introduction to Computational Psychology and AI

Spring 2026

Last updated: June 1, 2026

## Instructor

Zhixian Han

## Class Meeting Time and Location

## Office Hours

Alternative meeting times can be arranged if the regular office hours are not convenient.

## Course Description

This 3-credit course will provide an introduction to computational psychology and AI. The course will focus on using deep learning models and other mathematical and computational methods to build models that can help us understand the cognitive functions of the brain. The course will introduce some important computational models in lectures and will allow students to get hands-on programming experience in discussion sessions and homework assignments.

## Recommended Prerequisites

**Programming:** The course will primarily use Python for programming. Some programming experience with Python or any other programming language is highly recommended. Students who do not have any programming experience may still take the course if they are willing to learn and complete some preparatory tutorials at the beginning of the course.

**Math:** Basic knowledge about Probability and Statistics, Calculus, and Linear Algebra is highly recommended. Students who do not have sufficient math background may still take the course if they are willing to learn and complete some preparatory tutorials at the beginning of the course.

Psychology and Neuroscience: Knowledge about Psychology and Neuroscience is highly recommended.

List of recommended prerequisites:  
PSY 120000 and PSY 20100 or equivalent

## Tentative Outline

Week 1: Model Types  
Week 2: Python Workshop  
Week 3: Probability and Statistics  
Week 4: Linear Algebra and Calculus  
Week 5: Introduction to Reinforcement Learning  
Week 6: Reinforcement Learning and the Brain  
Week 7: Introduction to Artificial Neural Networks  
Week 8: Convolutional Neural Network  
Week 9: Hopfield Network  
Week 10: Spring Break (No Class)  
Week 11: Boltzmann Machine  
Week 12: Helmholtz Machine and Variational Autoencoder (**project proposal due by the end of this week**)  
Week 13: Attention, Transformer, and Natural Language Processing  
Week 14: Hidden Markov Model, Active Inference, and the Free Energy Principle Theory in Neuroscience  
Week 15: Project Presentations  
Week 16: Project Time (No Class) (**project final report due by the end of this week**)

For tutorials in weeks 1, 2, 3, 4, 5, 6, 7, 8, 12, 13, we adapted materials from Neuro-match Academy's Computational Neuroscience, Deep Learning, and NeuroAI courses under a CC-BY 4.0 license. In our version, we modified some content and added additional models and explanations to better fit our students' background. Full materials and original license: <https://neuromatch.io/open-education-resources/>.

## Course Format and Student Expectations

The class will meet twice a week, and students are expected to attend every class.

On Tuesday, the instructor will give a lecture to the class introducing the topic. The Thursday class meeting will be a discussion session in which students will discuss materials in the tutorials provided by the instructor. All tutorials are Jupyter Notebooks in Python, designed to be run on Google Colab.

## Discussion on Piazza

Starting in the second week, each student is required to post at least one discussion question or comment on Piazza, based on the current week's tutorial, by Wednesday at 7:00 p.m. If you have figured out the answers to your own questions after class, please post your answers below your questions on Piazza by next Monday. If you are still having difficulty, please explain why you are confused in a follow-up post by next Monday so that we can help you. These contributions will count toward your attendance/participation score. **You are also encouraged to answer your classmates' questions. Up to 5% extra credit may be earned by actively participating in discussions either in class or online on Piazza.**

(piazza sign up link: <https://piazza.com/purdue/spring2026/psy39200cpai>)

## Discussion session on Thursdays

Starting in the second week, one student will lead each discussion session on Thursday, and all students are expected to ask and answer questions. Each student should bring up the question they asked online and elaborate it for discussion during the discussion session on Thursday. **Please use the sign-up sheet on Brightspace to indicate the week in which you would like to serve as the discussion leader.** Depending on the total enrollment, one or two students will serve as discussion leaders each week. If a week has two discussion leaders, one will lead the first part of the tutorial discussion and the other will lead the second part.

Discussion leaders should post answers on Piazza after class to any online questions for that week that have not yet been answered online. If a discussion leader has difficulty answering a question, they should explain what they find confusing in a follow-up post so that the instructor can provide help.

The instructor will provide guidance during the discussion and post answers to some important questions on Piazza. There will also be some simple programming exercises for the students to complete in the tutorials. Students can discuss and complete the programming exercises during the class.

If any programming exercises are not finished during the class, students can complete them at home. **Students need to submit their completed tutorials by the end of next Monday.**

## Course Project

There will be no exams in this course; instead, students will complete a course project. Students may work independently or complete the course project in groups of two or three. The expected effort and output will be proportional to the group size.

For the course project, there are two choices:

1. Students can build a simple new computational model that is related to any psychology, neuroscience, or AI topics based on their interests.

2. Students can write a perspective article of any computational models that are related to any psychology, neuroscience, or AI topics based on their interests.

(A perspective article is an article that presents the author's viewpoint, interpretation, or vision on a specific issue, concept, or emerging direction in a field. Perspective articles are opinion-driven and forward-looking. They often highlight unresolved questions, propose new frameworks, or challenge prevailing theories, with the goal of stimulating discussion and inspiring future research. )

More specific instructions about the final project will be available later.

**Students are required to write a project proposal, give a project presentation (week 15), and write a final project report.**

## Extra Credit

You may earn extra credit (up to 5%) if you actively participate in the discussion in class or online on Piazza (e.g. provide good answers to other students' questions).

## Late Policy

If you submit your work late, there will be no penalty as long as it is submitted before I begin grading. A 20% penalty will be applied if the submission is made after the I have started grading. No points will be awarded if the work is submitted after the grades for the corresponding assignment have been released.

If you have a legitimate reason for submitting an assignment late, please notify me in advance by email so that an exception can be made.

## Incorrect Submissions

If you submit a homework assignment, project proposal, or project report in an incorrect format (e.g., not a PDF) and we are unable to open the file, you will be allowed one resubmission with a 20% penalty. Any further incorrect submissions will receive 0 points.

## Educational Platforms

### Google Colab:

We will primarily use Google Colab to run the code in the tutorials. The tutorials are Jupyter Notebooks in Python, so it is also possible to run the code in other environments if using Colab is not convenient for the student.

### Brightspace:

The course materials, including lecture notes, tutorials, assignment announcements, and grades will be posted on Brightspace. The weekly assignments (completed tutorials) should be submitted on Brightspace.

**Piazza:**

We will use Piazza for class discussion.

(sign up link: <https://piazza.com/purdue/spring2026/psy39200cpai>)

We also encourage you to post them on Piazza if you have any other questions about the course content, assignments, or project. It is a shared discussion forum, where you can get help quickly and efficiently from your classmates and the instructor while benefiting other students with similar questions.

## Assessment and Grading

Final Grades will be calculated as follows:

Attendance/Participation: 30%

Leading Discussion Sessions: 10%

Completing Weekly Tutorials: 40%

Final Project: 20%

Extra Credit: 0% - 5%

Final Grade Cutoffs:

90%: A

80%: B

70%: C

60%: D

less than 60%: F

+ and - Grades will be assigned to the scores close to the above cutoffs.

## Academic Honesty

You are encouraged to discuss the exercises in tutorials and other course materials with other people. However, you are expected to complete all tutorial assignments by yourself after you have understood the materials.

For the course project, you are expected to develop models, implement the models with code, and write a project report by yourselves. You may get help from outside resources when necessary, but you must be the primary individuals conducting the project, understand every part of it, and ensure that your course project reflects your own ideas.

## **Nondiscrimination Statement**

Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her own potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life. More details are available on our course Brightspace table of contents, under University Policies.

## **Accessibility**

Purdue University strives to make learning experiences as accessible as possible. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let me know so that we can discuss options. You are also encouraged to contact the Disability Resource Center at: [drc@purdue.edu](mailto:drc@purdue.edu) or by phone: 765-494-1247. More details are available on our course Brightspace under Accessibility Information.

## **Mental Health Statement**

If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try WellTrack. Sign in and find information and tools at your fingertips, available to you at any time.

If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M-F, 8 am - 5 pm. If you find yourself struggling to find a healthy balance between academics, social life, stress, etc. sign up for free one-on-one virtual or in-person sessions with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is completely free and can be done on BoilerConnect. If you have any questions, please contact Purdue Wellness at [evans240@purdue.edu](mailto:evans240@purdue.edu).

If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS office of the second floor of the Purdue University Student Health Center (PUSH) during business hours.

## **Disclaimer**

This syllabus is tentative and subject to change. Any substantive change will be communicated to the students.